DATA ITEM DESCRIPTION

Title: HIGH-ALTITUDE EMP (HEMP) HARDNESS MAINTENANCE AND HARDNESS SURVEILLANCE (HM/HS) PLAN

Number: DI-EMCS-81855A
AMSC Number: 10416
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20230320
Limitation: N/A
GIDEP Applicable: No
Project Number: EMCS-2023-011

Use/Relationship:

The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan applies to military standard MIL-STD-3023A, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Military Aircraft*. DI-EMCS-81855A *High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan*, is hereby cancelled and superseded by DI-EMCS-81855A. This data item description (DID) contains the format, content, and intended-use information for the data deliverable resulting from the HM/HS tasks described in sections 4.8 and 5.8 and Appendix E of MIL-STD-3023A, as applicable.

The purpose of the HM/HS program is to ensure that the HEMP-protection subsystem's (HPS') hardness levels are preserved throughout the service life of the aircraft. HM/HS procedures comprise the preventive maintenance, inspection, test, and repair activities accomplished on a HEMP protected operational aircraft.

The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan serves to describe each HEMP hardness-critical item (HCI) and assembly (HCA) in the aircraft, the specification for that item/assembly, and the tests required to verify its performance. The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan provides the means for the government to ensure that government prescribed methods and procedures are used by aircraft maintainers and that fielded aircraft undergoing planned modifications and subject to normal wear and tear remain HEMP-survivable over their life-cycle.

The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan provides the data and information for the Government to evaluate the continued compliance of the aircraft with its HEMP-survivability requirements throughout its service life, and to manage any planned modifications to the aircraft or its systems.

The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan identifies the data acquisition equipment; frequency clearance requirements and deviations, if required; and the approval and site notification process.

The Department of Defense's HEMP-certification authority requires that a funded HM/HS program and a documented High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan be submitted as part of a HEMP-certification request package for a facility or system.

This DID is normally applied immediately after aircraft are delivered to a user command.

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This DID is related to DI-EMCS-81847A, DI-EMCS-81849A, DI-EMCS-81850A, DI-EMCS-81852A, and DI-EMCS-81854A.

This DID supersedes DI-EMCS-81855.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. HM/HS program documentation shall be maintained over the life cycle of the aircraft to include all calibration, hardness assurance and verification test data. The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan's classification shall be in accordance with any relevant aircraft- or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities. (DTRA's security classification guide is available by mail request to ATTN: RD-NTSA, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA 22060-6201.)

2. Format. Contractor format is acceptable.

3. Content. This DID describes the overall High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan, which describes the HM and inspection procedures, HS test procedures, re-verification test procedures, supply support and training requirements, and technical data requirements for implementing the HM/HS program. The test methods are written to be applicable at varying levels of maintenance including depot, intermediate, and operational levels.

3.1 Administrative data. The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan shall contain an administrative section covering the following:

a. Contract number, CDRL number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan

3.2 Summary information. The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan shall summarize the following:

3.2.1 Introduction, to include:

a. General objectives of the HM/HS program and its procedures

b. A general description of the HPS' topological hardness to be maintained and to undergo surveillance

c. Statements of any assumptions or limitations associated with aircraft HM/HS inspection and test procedures, including for subsystems

d. Statements of any assumptions or limitations associated with aircraft HPS performance tests and HEMP verification tests, including for immunity testing of all MCS

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e. A listing of sources for baseline test data derived from the aircraft or system's hardness assurance, HPS performance, and HEMP verification testing

f. Descriptions of any issues regarding the HPS

g. A description of all shield monitoring capabilities, including test procedures

h. References to all operational and maintenance manuals for the aircraft and its mission critical systems (MCS)

3.2.2 Scope. A general description of the overall HM/HS requirements matrix that is used to maintain and demonstrate the HPS' sustained lifecycle compliance with MIL-STD-3023A's performance requirements. This description will include the relative roles of routine maintenance procedures (i.e., HM), and of analyses, tests and inspections (i.e., HS).

A logistics support analysis shall be described to define the HM/HS requirements, supply support requirements, training requirements, and technical data requirements, and identify all HCIs and HCAs. A means for maintaining configuration management as the HPS, including associated MCS special protective measures, is modified will also be described.

3.2.3 Methods of HM/HS. A general description of the procedures that are used to confirm the aircraft or system operator's compliance with each HM/HS requirement as listed in section 3.3 below.

3.2.4 Engineering factors. Any important engineering factors affecting the HM/HS procedures, such as: physical-accessibility constraints of the aircraft's or system's HPS topology, resource limitations, safety and security protocols, and precedent reports.

3.3 Detailed information. The High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan shall describe in detail at least the following information:

a. One-line wiring diagrams for power sources

b. Comprehensive, aircraft- or system-specific HM procedures for maintaining the hardening status of the HPS with emphasis on HCIs

c. Comprehensive, aircraft- or system-specific HS test plans for surveillance testing of the HPS for determining hardening status

d. Supply support requirements for both HM and HS

These components may be combined in a single document, or they may be separated into multiple, cross-referenced documents.

The documents shall also contain the following detailed information:

3.3.1 HM and inspection procedures. Detailed procedures for preventive maintenance, inspection, repair and replacement for the aircraft HEMP hardness critical assemblies or items and associated MCS shall be described, and all HCIs/HCAs shall be identified. Candidate maintenance test procedures are described in MIL-STD-3023A, Appendix E. These procedures shall include the maintenance of a data log. A schedule shall be identified for how often the outlined procedures are to be performed. The descriptions of HM procedures should be linked to the descriptions of the

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aircraft's or system's HPS topology. Where applicable, pass/fail criteria should be relatable to production hardness pass/fail criteria derived from the hardness allocation procedure (MIL-STD-3023A, section 5.2 and Appendix A).

3.3.2 HS test procedures. Detailed HS test procedures for the aircraft and its MCS shall be developed and used. These procedures shall include the maintenance of a data log. A schedule shall be identified for how often the outlined procedures are to be performed. The HS procedures shall be designed to identify HPS degradations that may have occurred during operation, and to provide assurance that any flaws are rectified during programmed depot or emergency maintenance to thereby assure HEMP-hardened aircraft are returned to service after programmed depot maintenance. These procedures shall be applied to the HPS' EM shields, penetration protective devices (PPDs) and special protective measures (SPMs).

The HS data should be relatable to the pass and fail criteria used in the HEMP HPS performance and verification tests. The data from these tests will be used as a baseline for HS data comparison.

3.3.3 Reverification test procedures. The reverification test procedures shall utilize the baseline verification test data of the first aircraft in each fleet. Subsequent re-verification test sample sizes shall be dependent on mission category or aircraft type. The reverification test procedures shall be the same as the system verification test procedures (see MIL-STD-3023A, 5.7.3 and Appendix D).

3.3.4 Supply support requirements. A parts list of the installed HCIs and lists of recommended aircraft or system HPS's spare parts, repair parts, supplies, special tools, and special test equipment shall be provided.

3.3.5 Training requirements. Training requirements for personnel performing aircraft surveillance testing, maintenance, upgrades and repairs shall be identified. Identification of the type of training manuals to be prepared should be identified.

3.3.6 Technical data. Aircraft data logs for HM and HS will be maintained in chronological order. A configuration management log will also be maintained in chronological order showing all changes to the aircraft's HPS topology, e.g., low-risk electromagnetic (EM) barriers, HCIs, and MCS.

Also, an HPS technical manual shall be provided to contain, at a minimum, the following:

a. HPS description and principles of operation

b. Maintenance and inspection procedures repair and replacement procedures and HS test procedures

c. Supply support requirements

d. Training requirements

3.4 Other information sources. When other available information sources contain data required by the High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan, these sources shall be referenced, rather than duplicated, within this High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan.

End of DI-EMCS-81855A

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